

A Comprehensive Survey on Credit Card Fraud Detection using Advanced Deep Learning

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Abstract

Due to the increase in online transactions and sophisticated cybercrime, many critical challenges are evolving in recent time. One such challenge is credit card fraud detection. This survey reviews conventional methods for real-time fraud detection, focusing on models: Deep Neural Networks, Convolutional Neural Networks, and Long Short-Term Memory networks. It analyzes complex transactional patterns and user behavior. Techniques such as one-hot encoding for categorical data and Generative Adversarial Networks for addressing class imbalance are also examined. We evaluate methods for entering Random Forest classifiers and attention mechanisms to improve detection accuracy. By analyzing performance metrics like recall, precision, Area under the Curve and F1-score, this paper assesses the effectiveness of models in real-time scenarios. Additionally, we discuss the scalability of these solutions for live financial systems, providing a comprehensive overview of advanced techniques for credit card fraud detection. This survey highlights significant financial impact due to credit card fraud and reviews optimized deep event-based network frameworks for detection. We compare various deep learning methodologies, their approaches, accuracy, and findings. We also demonstrate the superiority of deep learning techniques over conventional machine learning techniques to manage imbalanced datasets and enhancing detection accuracy.

Keywords: Deep Learning, Machine Learning, Credit Card Fraud Detection, CNNs, LSTMs, GANs, Data Augmentation, Real-time Detection.

1. Introduction

The surge in online transactions has significantly increased credit card fraud, necessitating advanced detection methods to protect financial systems and consumers. Traditional rule-based systems fall short, prompting the development of an Optimized Deep Event-Based Network Framework using deep learning to improve fraud detection. This framework includes several key components: data preprocessing (cleaning, normalizing, and transforming raw data), feature extraction (identifying and extracting relevant attributes), model training (to learn data patterns, Machine Learning (ML) algorithms is applied), and optimization techniques (fine-tuning model parameters for better performance).

The framework aims to accurately detect fraud in real-time, even with imbalanced datasets where fraudulent transactions are rare. Techniques like oversampling, under sampling, and synthetic data generation help balance the dataset, enhancing the model's ability to identify fraud without compromising on legitimate transactions. ML and Deep Learning (DL) offer advantages over conventional techniques by autonomously learning large datasets and

adapting to new patterns in real time, crucial in dynamic landscape of financial fraud. This survey paper analyzes recent advancements in Credit Card Fraud Detection (CCFD) using ML and DL, focusing on state-of-the-art models with exceptional performance [11]. A recurrent neural network namely Long Short-Term Memory networks (LSTM) can effectively capture the temporal dependencies in the sequential transaction data, detecting complex fraud patterns over time. Convolutional Neural Networks (CNNs) [10], is adapted to detect fraud case by treating transaction data as two-dimensional input, identifying spatial patterns indicative of fraudulent behavior. Attention mechanisms, inspired by human cognition, enhance deep learning architectures' focus on relevant data aspects, improving fraud detection accuracy. The survey addresses the imbalance issues confronted in CCFD, highlighting Generative Adversarial Networks (GANs) for data augmentation. GANs generate synthetic fraudulent transactions resembling real cases, increasing minority class representation in training data and improving model performance in identifying rare fraud patterns. Hybrid approaches [6-8] integrating traditional machine learning with deep learning enhance fraud detection. These systems combine classical machine learning algorithms' interpretability and efficiency with deep neural networks' feature learning capabilities. Ensemble methods combining predictions from multiple models, including traditional and deep learning approaches, show superior performance, leveraging model diversity. This survey provides insights into the current state of CCFD using advanced ML or DL techniques, highlighting strengths and limitations and guiding future research in combating financial fraud.

2. Proposed Methods

2.1 Background

Traditional fraud detection ML methods like Logistic Regression (LR), Random Forest (RF), Support Vector Machines (SVM), Decision Trees (DT) and rule-based systems often face limitations in handling the dynamic and high-dimensional nature of credit card fraud. These approaches struggle with class imbalance and tend to generate high false-positive rates. In contrast, DL models, like transformers, CNNs, LSTMs, excellently capture the complex sequential patterns of the transactional data. Particularly, event-based recurrent networks and transformers process transactions as event sequences, revealing temporal dependencies indicative of fraud. Optimization techniques like hyper parameter tuning and ensemble methods further boost their accuracy and lower false-positive rates, marking a substantial improvement over traditional method.

2.2 Literature Survey

The research papers on CCFD relies on advanced ML or DL techniques to address the evolving and complex nature of fraudulent transactions.

Table 1. Summary of Research Paper Reviewed

No.	Work	Algorithms	Features	Data Traces	Traffic Considered	Classification Level
1	Critical Infrastructure investigation using optimal stacked auto encoder [1]	MRFO, OSAE, COA	- Packet analysis using Ettercap, Wireshark, and TCP dump - Feature selection using MRFO - Classification via OSAE - Parameter tuning with COA	Bot-IoT and UNSW_NB15 datasets	Industrial network security anomalies	Fine-grained detection

2	Deep Ensemble Algorithm Framework for CCFD [2]	Extra-Tree Ensemble, Adam Optimization	- Deep learning with ensemble techniques - Adaptive optimization for fraud prediction - Feature transformation into tensors	Credit card transaction dataset (284,807 total, 492 fraud cases)	Financial transactions	Fine-grained (fraud detection)
3	Review on deep learning for CCFD [3]	CNN, RNN, LSTM, GRU	- Compares architectures like MLP, CNN, RNN, LSTM, GRU, BiLSTM, Transformer - Resampling, ensemble transfer and learning techniques	Real-world CCFD datasets	Financial transactions	Fine-grained
4	Modeling and Learning Strategies of CCFD [4]	Neural Networks, SVM, LR, Random Forests, DT, and Anomaly Detection	- Classifiers analyze authorized transactions and flag suspicious ones - Investigates verification latency and alert-feedback mechanisms	Large real-world dataset (75M transactions)	Financial transactions	Fine-grained
5	Deep Learning Approaches for CCFD [5]	MLP, CNN, SMOTE	- Handles class imbalance with SMOTE - Multi-layer neural network	Not mentioned	Financial transactions	Fine-grained
6	Data Resampling and DL Ensemble for CCFD [6]	GRU, MLP, SMOTE- ENN, LSTM	- SMOTE-ENN resampling improves classification accuracy - Stacking Ensemble model with MLP as meta-learner	Credit card fraud dataset	Financial transactions	Fine-grained
7	Framework for Early Detection of Pushing in Crowds [14]	EfficientNetV2 B0, Optical Flow Model	- Uses live video analysis - Integrates deep learning for behavior detection	Real-world event entrance dataset	Crowd movement behavior	Fine-grained
8	Deep Neural Networks based CCFD [8]	Deep Neural Network (7-layer architecture)	- Merges identity and transaction datasets - Uses label encoding and SMOTE	IEEE-CIS Fraud Detection dataset	Financial transactions	Fine-grained
9	Novel Framework for CCFD [9]	Random Forest, LR, DT, KNN, SVM, ANN, SVDD, PSLPSO	- Feature selection techniques (filter, wrapper, embedded) - SVDD for classification	Not mentioned	Financial transactions	Fine-grained

10	ML and DL Algorithms based CCFD [10]	CNN, DBN, Autoencoder, LSTM, RF, DT, SVM, LR, NB	20-layer CNN model - PCA for dimensionality reduction	Credit card fraud dataset	Financial transactions	Fine-grained
11	DL based CCFD [13]	GRU, LSTM, RNN, ANN	- Uses 80M transactions - Handles missing data and class imbalance	80 million transactions	Financial transactions	Fine-grained

2.2.1 DL in Fraud Detection

This section delves into the advancements and applications of DL in CCFD. It highlights DL abilities to address patterns even in high-dimensional data. As a specialized branch of ML, DL leverages deep neural networks (DNNs). DNN is a multi-layered neural networks. It helps in learning complex relationships by varying neuron weights through gradient descent. Neurons, connected by these weighted links, sum incoming data and apply non-linear activation functions to yield outputs, minimizing discrepancies between predictions and actual outcomes through repeated adjustments. DNNs, characterized by their multiple hidden layers, excel over shallow ML methods for complex, large-scale tasks like fraud detection, where data imbalance, scalability, and real-time processing demands are critical challenges. Within DL, various architectures have proven highly effective for specific data types and fraud detection tasks. CNNs, for example, are proficient in recognizing spatial hierarchies of transactional data [10]. This reveals fraud patterns otherwise undetectable by traditional methods. Recurrent Neural Networks (RNNs), in contrast, specialize in sequential data, making them ideal for modeling transaction sequences over time. However, standard RNNs are challenged by gradient vanishing or exploding issues, which limits learning from long-term dependencies. Gated Recurrent Units (GRUs) and LSTM network were introduced to mitigate these issues. LSTMs manage long-range dependencies through specialized memory cells, while GRUs simplify the architecture by combining memory and input functions, preserving essential sequence information [11]. Bidirectional versions of RNNs, LSTMs, and GRUs further enhance prediction accuracy by analyzing both past and future contexts within transaction sequences. This bidirectional approach for identifying subtle signs of fraud in sequences of credit card transactions is effective. Meanwhile, hybrid DL frameworks, which combine multiple DL architectures, have demonstrated exceptional performance in capturing intricate fraud patterns and reducing false alarms. These hybrid approaches often integrate techniques to overcome the challenges encountered in detection of fraud scenarios like data imbalance and real-time processing

2.2.2 Key Techniques for Enhancing Detection Accuracy

A variety of DL models, including CNNs, LSTM [11], and GRU, have demonstrated efficacy in detecting fraudulent transactions by capturing sequential and temporal patterns. Ensemble frameworks that combine these architectures with multilayer perceptron (MLPs) as meta-learners show enhanced performance by leveraging the strengths of individual models. Stacking LSTM [11] and GRU in ensemble configurations improves model sensitivity and specificity, handling the intricacies of fraud detection more effectively than single DL models.

Multi Layered Perceptron (MLP):

The Multi Layered Perceptron [10] (MLP) processes transaction data through layers of neurons. It has three layers namely: input, multi hidden and output layers. At each layer, every neuron calculate weighted sum the inputs from the previous layer. Non-linearity is introduced by adding bias and activation function.

$$Z^l = W^l A^{l-1} + b^l \quad (1)$$

The Rectified Linear Unit (ReLU) is an activation function, which is given as:

$$A^L = \sigma(4(L)) \quad (2)$$

where $\sigma(x) = \max(0, x)$

This structure helps MLP capture complex, non-linear relationships within data, improving fraud detection accuracy. The output layer often uses a softmax function for classification:

$$\hat{y} = \text{softmax}(Z^L) = \frac{e^{Z^L}}{\sum e^{Z^L}} \quad (3)$$

Convolutional Neural Networks:

CNNs excel at recognizing spatial patterns in transaction data, treating the data as a 2D grid. Each convolution layer applies a filter (kernel) over the data, generating feature maps:

$$Z^{(l)} = \sigma \left(\sum_c W_c^{(l)} * A_c^{(l-1)} + b^l \right) \quad (4)$$

The ReLU introduces non-linearity, helping detect subtle, localized patterns in transaction sequences that may indicate fraudulent behavior. A pooling layer, typically max pooling, reduces the feature map's size:

$$A^l = \max_{i,j} (Z^{(l)})_{ij} \quad (5)$$

Long Short-Term Memory networks:

LSTM are particularly used for sequential data handling. These are ideal for tracking transaction histories over time. They maintain a cell state, which updates selectively through different gates:

Forget Gate:

$$f_t = \sigma(W_f * [h_{t-1}, x_t] + b_f) \quad (6)$$

Input Gate:

$$i_t = \sigma(W_i * [h_{t-1}, x_t] + b_i) \quad (7)$$

Cell State Update:

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \quad (8)$$

Output Gate:

$$o_t = \sigma(W_o * [h_{t-1}, x_t] + b_o) \quad (9)$$

Hidden State Output:

$$h_t = o_t \odot \tan C_t \quad (10)$$

These equations allow LSTM [11] to retain important historical transaction patterns while discarding irrelevant ones, helping identify abnormal behavior sequences. LSTM [11] maintains a cell state and controls information flow using gates:

- Forget Gate → It decides, what is to be discarded.
- Input Gate → It determines, what new information is to be stored.
- Output Gate → It controls output from the memory cells.

Gated Recurrent Unit (GRU [11]):

GRU is a simpler alternative to LSTM [11] that removes the **cell state** and uses only two gates. Instead of three gates, GRU has:

- Reset Gate → It controls amount of past information to be forgotten.
- Update Gate → It decides amount of past information to be retained.
- Final Hidden State Updates

Transformer- [11] [10] based Model (TabTransformer):

Transformers [11] [10], particularly TabTransformer [10] for tabular data, weight the various sections of input sequence using self-attention. Due to this, relevant features can be effectively observed in transaction data. This method of self-attention computes a weighted combination of values based on key, query and value matrices:

- Self-Attention Mechanism:

$$\text{Attention}(Q,K,V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (11)$$

Where, Q is query, K is key and V is value.

- Multi-Head Attention:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O \quad (12)$$

- Feedforward Network:

$$\text{FFN}(x) = \text{ReLU}(xW_1 + b_1)_2 + b_2 \quad (13)$$

Application in Fraud Detection:

- Works with large, complex datasets.
- Automates feature extraction from categorical and numerical variables.

Autoencoders [12]:

Autoencoders [12] are unsupervised networks that reconstruct input data and flag anomalies.

- The encoder compresses input data.
- The decoder reconstructs it.
- If the reconstruction error is high, the input is likely fraudulent.

Application in Fraud Detection: Detects outliers (fraudulent transactions) with high reconstruction errors.

2.2.3 Feature Selection and Dimensionality Reduction

Given the high dimensionality of transaction data, several studies have employed feature based selection techniques. It retain relevant features of model that is trained. Methods like Support Vector Data Description (SVDD) [5] and Particle Swarm Optimization (PSO) enhance model efficiency by reducing computational costs without compromising accuracy. The use of deep autoencoders further aids in feature extraction, mapping transaction data to higher-dimensional spaces to retain essential transaction patterns and characteristics

2.2.4 Real-Time Processing and Scalability:

The application of frameworks such as DEAL [2] (Deep Ensemble Algorithm) and cloud computing platforms, utilizing Google TensorFlow and Keras, enables real-time fraud detection and scalability. These frameworks allow fraud detection models to process large transaction volumes quickly, making them suitable for deployment in live financial systems. Techniques like adaptive optimization within ensemble DL frameworks help adapt to new fraud patterns, demonstrating potential for practical application in high-frequency transaction environments.

2.2.5 Performance Evaluation Metrics:

The performance metrics commonly used in fraud detection models are – F1-score, precision, F1-score, AUC and recall. Studies consistently indicate that DL models, particularly those using CNNs, achieve higher AUC and accuracy than traditional ML models. Advanced DL approaches achieve improved precision and recall by focusing on hard-to-classify fraud cases and minimizing false positives [11]. High AUC scores, often reaching around 99 percent, underscore the effectiveness of DNNs in differentiating between fraudulent and legitimate transactions.

3. Results & Discussion

3.1 Methodology

A structured methodology for CCFD is employed using ML and DL models. The approach begins with **data collection and preprocessing**, where key transaction attributes such as transaction amount, merchant category, location, time, and fraud labels are cleaned, normalized, and transformed to enhance fraud detection performance.

Various fraud detection models are utilized, including traditional ML techniques such as LR, DT, Random Forest, and SVM, alongside advanced DL models like MLP, LSTM, CNNs, Transformers, and GANs. Additionally, hybrid and ensemble approaches, including Stacking Ensembles, Autoencoders [12], and Graph Neural Networks (GNNs), enhance detection accuracy by combining multiple model outputs.

To further optimize model performance, data balancing method like SMOTE [6], ENN, Random Under sampling are applied for class imbalance handling. At the same time, optimization algorithms like Adam, RMSprop, and SGD [10] are used for model tuning. These all models are evaluated on the grounds of following metrics: accuracy, F1-

score, precision, recall and ROC-AUC [11] to ensure reliable fraud detection.

A comparative analysis indicates that DL models consistently outperform conventional ML models. The performance in sequential fraud detection and recognizing complex fraud patterns is better than other applications. Stacking Ensembles [6] achieve the highest accuracy and recall, demonstrating the effectiveness of integrating multiple models for fraud detection. This methodology ensures scalable as well as adaptive fraud detection that is able to evolve along the emerging fraud patterns in the real-world financial applications.

3.2 Comparative Analysis

A comparative analysis of various DL models and frameworks reveals that DL techniques generally outperform conventional ML techniques in fraud detection. For instance, a study comparing ML with DL algorithms found that DL models achieved higher accuracy, precision, and AUC scores. Another study demonstrated that a deep learning ensemble with data resampling outperformed widely used ML classifiers.

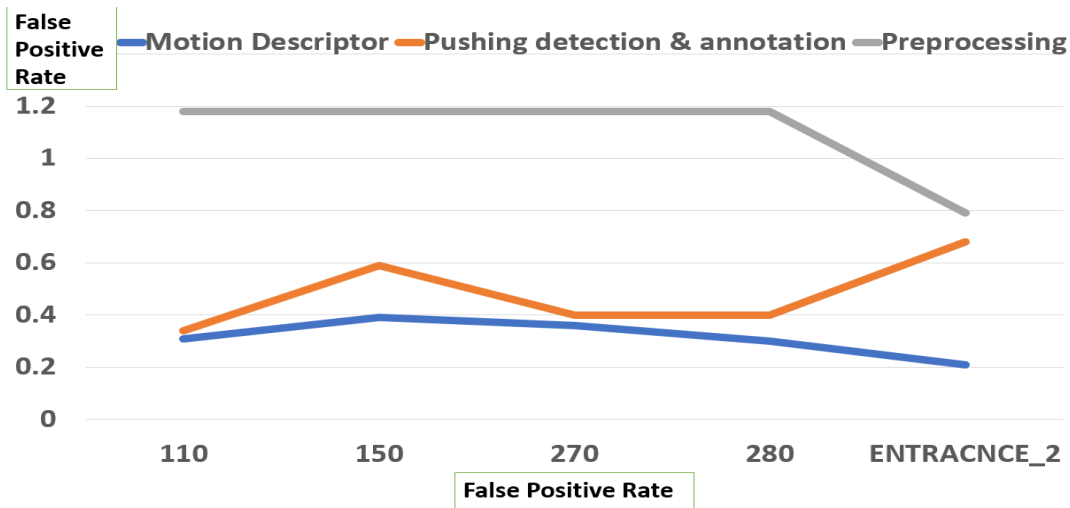


Fig.1. Weighted factor values for full features.

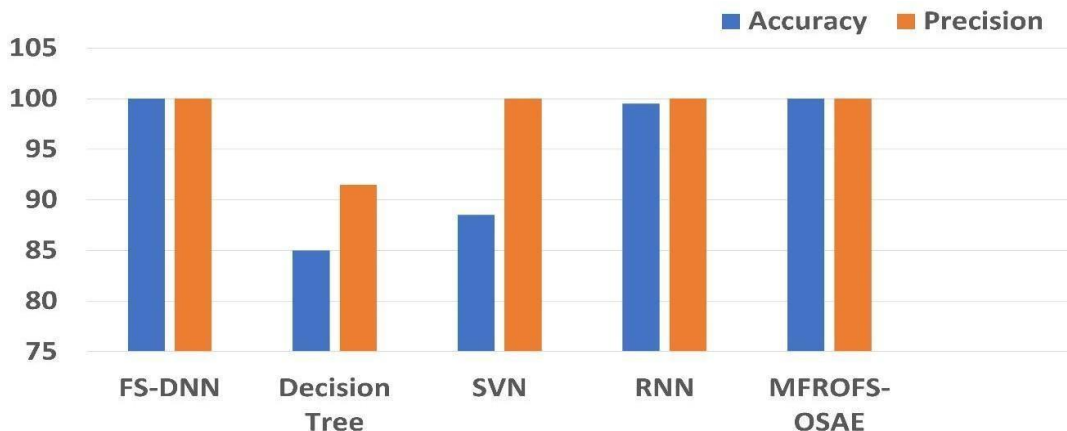


Fig. 2. Comparison of accuracy and precision

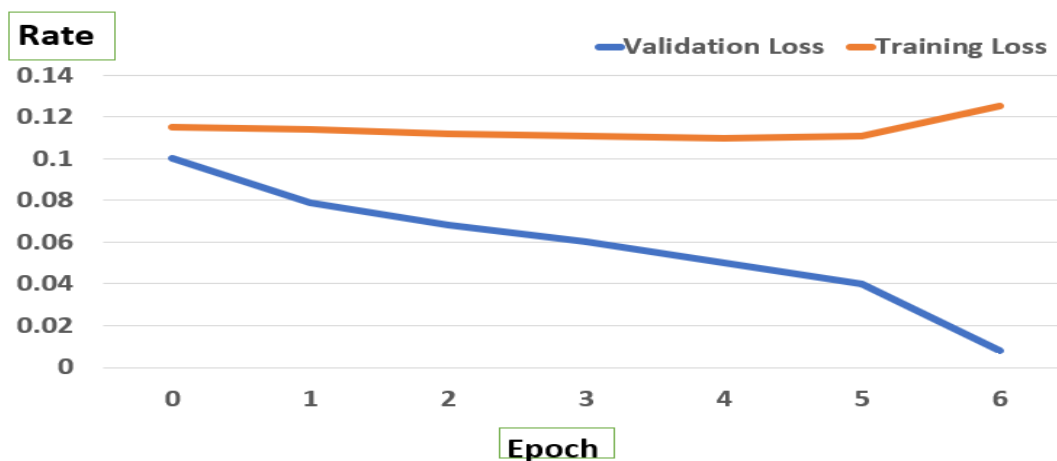


Fig. 3. Comparison of validation and training loss

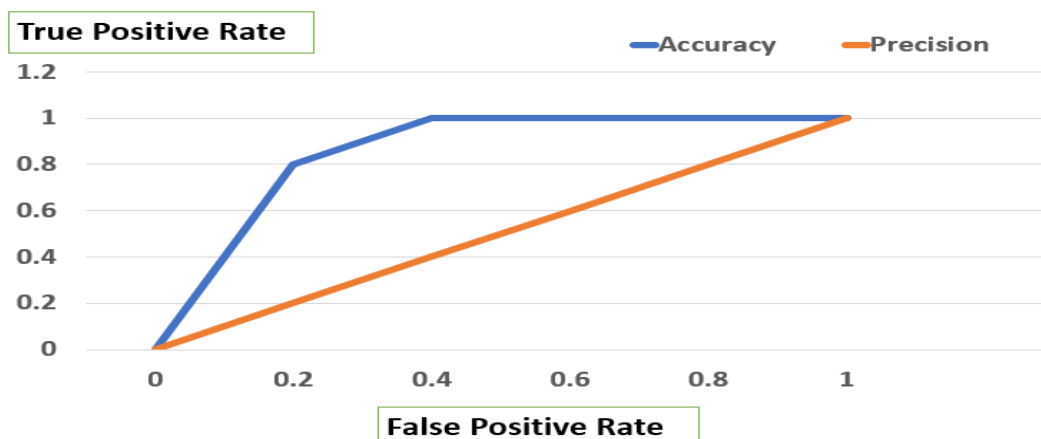


Fig. 4. Comparison of accuracy and precision

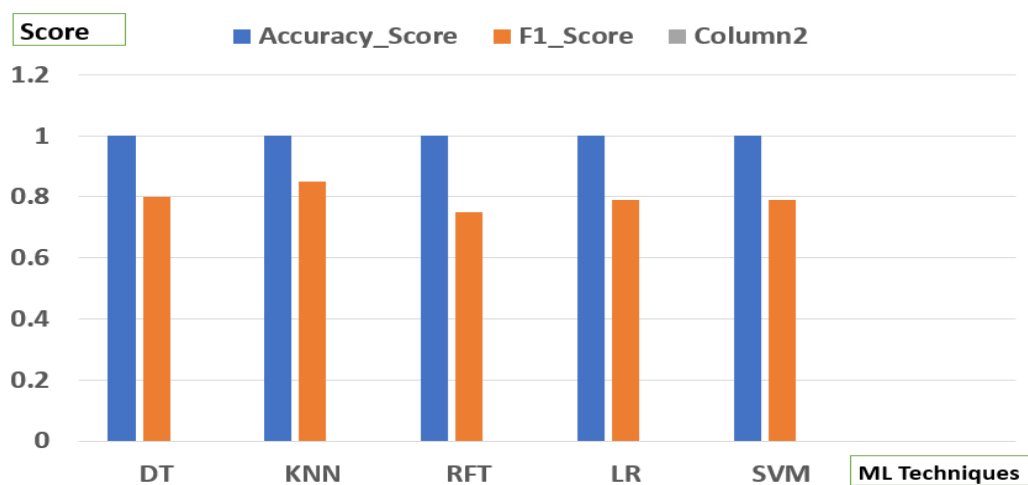


Fig. 5. Comparison of Accuracy and F1_score

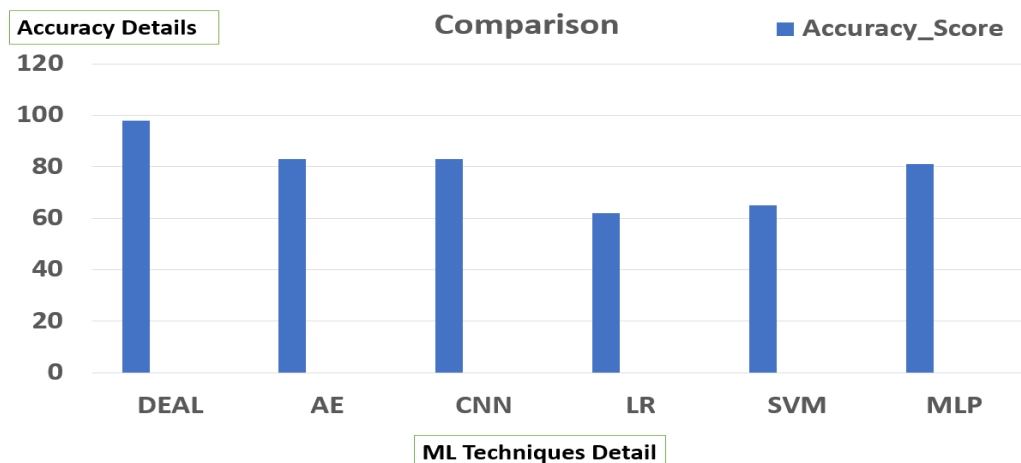


Fig. 6. Comparison of Accuracy score

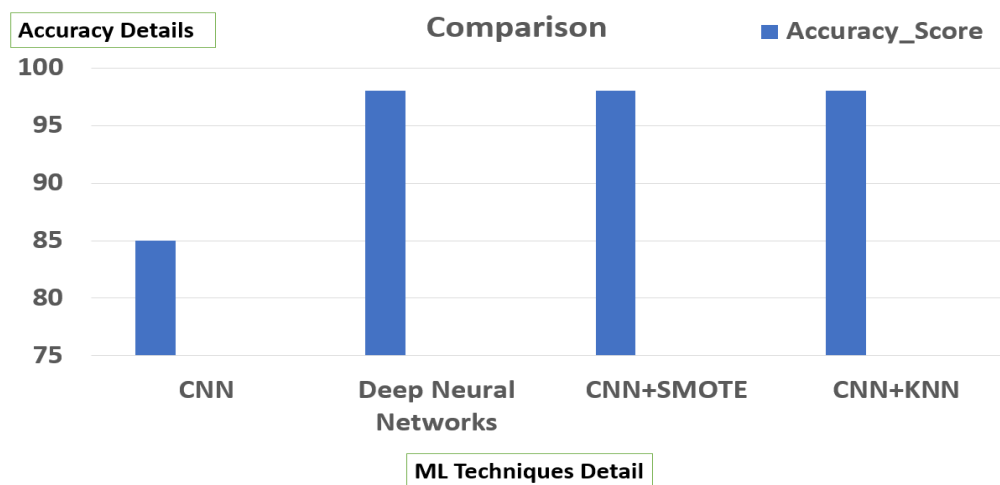


Fig. 7. Comparison of Accuracy_score

Displays the importance (weight) of various features used in fraud detection. Likely highlights that transaction amount, time, and location have higher influence in classification models. Compares different models (e.g., CNN, LSTM, RF) based on how accurate they are in general and how precise they are in identifying fraud cases. Shows the learning behavior of deep learning models. A small gap between training and validation loss suggests good generalization; large gaps may indicate overfitting. This appears similar to Fig. 2, possibly representing another experiment or extended dataset comparison. Again, highlights trade-offs between overall accuracy and fraud precision. Evaluates model balance: F1 score has effect on both - precision as well as recall, giving insight into how well a model balances false positives and false negatives. Directly compares the accuracy of multiple models. Likely emphasizes that deep learning models outperform traditional ones. A secondary or focused accuracy comparison, possibly for a subset of models or a specific data resampling technique (e.g., SMOTE vs. VAE).

4. Current Limitations and Future Directions

Despite the notable advancements, DL models still face challenges related to interpretability, computational complexity, and adaptation to evolving fraud tactics. The high dependency on labeled data for supervised learning models is a limiting factor, particularly in semi-supervised or unsupervised contexts. Future research is needed to

develop more adaptable models capable of handling real-time data. Additionally, exploring the interpretability of DL models remains essential for their acceptance and deployment in financial institutions.

5. Conclusion

This work surveys CCFD techniques using DL and ML models. Hybrid approaches like CNN +SMOTE, CNN +KNN, and ensemble frameworks achieved over 97% accuracy with high precision and recall, effectively handling imbalanced datasets. Class imbalance significantly affects performance, making precision and recall crucial. Oversampling methods like VAE and SMOTE improved fraud detection sensitivity but slightly lowered precision. Advanced architecture like GRU and LSTM effectively captured transaction patterns, reducing false positives, while the DEAL framework enhanced detection rates. Hybrid and ensemble models, particularly DEAL (98.2% accuracy, 97% precision) and CNN +KNN (98.0% accuracy, 96.2% precision), balanced accuracy and fraud detection effectiveness best. Traditional models like Logistic Regression and MLP underperformed. Future research intent to improve interpretability, real-time deployment, and efficiency while maintaining high detection rates.

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